

School of Data Science
ASSESSMENT FORM FOR DDA2081/2082/2083
Independent Study I/II/III

Basic Information:

Term	Academic Year 2025-26, Term 1
Course Code and Title	☒ DDA2081 Independent Study I ☐ DDA2082 Independent Study II ☐ DDA2083 Independent Study III
Student Name	王翼城 Yicheng Wang (SDS)
Student ID	123090586
Project Topic	Hardware/Software Co-Design of an FPGA-Accelerated Image Signal Processing Pipeline on a RISC-V SoC Platform
Supervisor	Yeh-Ching Chung

Guidance notes:

1. This Assessment Form will be used to record marks for each assessment component.
2. Each assessment component **should be marked out of 100**.
3. Below is the assessment structure for the UG course Independent Study.

Assessment	Weighting	Score	Grade
Presentation	40%	36	A
Report	60%	55	
	Total	91	

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- Comments made by supervisor (if any):
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The midterm report and the final report are well-organized and well-written.

1. In the midterm report, Yicheng has done the following things
 - Completed the software simulation and profiling on Windows, identifying the "Denoise" module as the critical bottleneck (consuming $\approx 60\%$ of CPU time).
 - Designed the heterogeneous system architecture based on these findings.
2. In the final report, Yicheng has done the following things
 - Completed the physical setup and operating system deployment of the BeagleV-Fire development board, preparing the ground for the upcoming hardware implementation.
 - Focuses on the theoretical study and architectural design of the FPGA-based denoise accelerator

Overall, this independent study established the theoretical and methodological foundation required for the FPGA implementation and system integration for Image Signal Processing Pipeline on a RISC-V SoC Platform.

Signature of Supervisor : 

Date : 2025.12.11